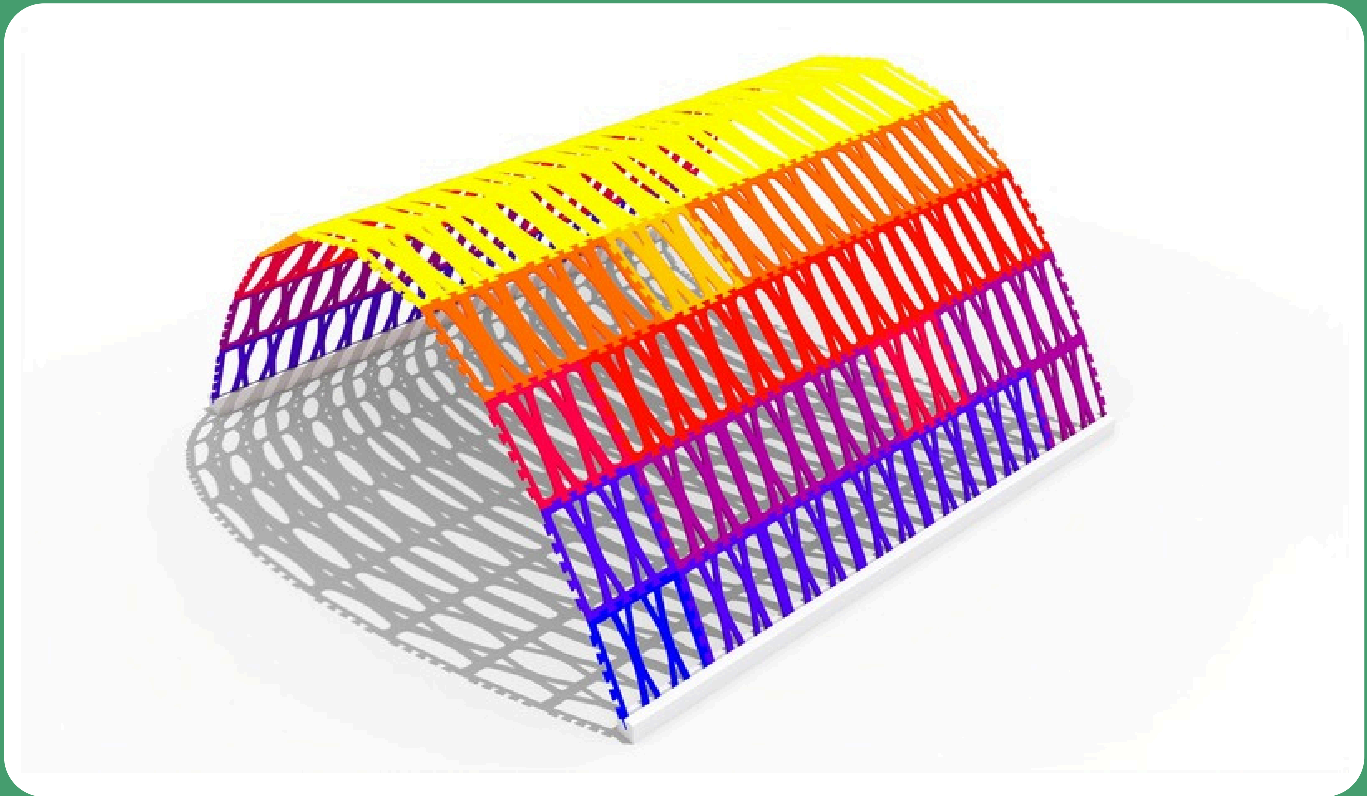
TOPOLOGY OPTIMIZATION FOR ARCHITECTURAL STRUCTURES



TYPE: Structural Optimization

YEAR: 2023-2024

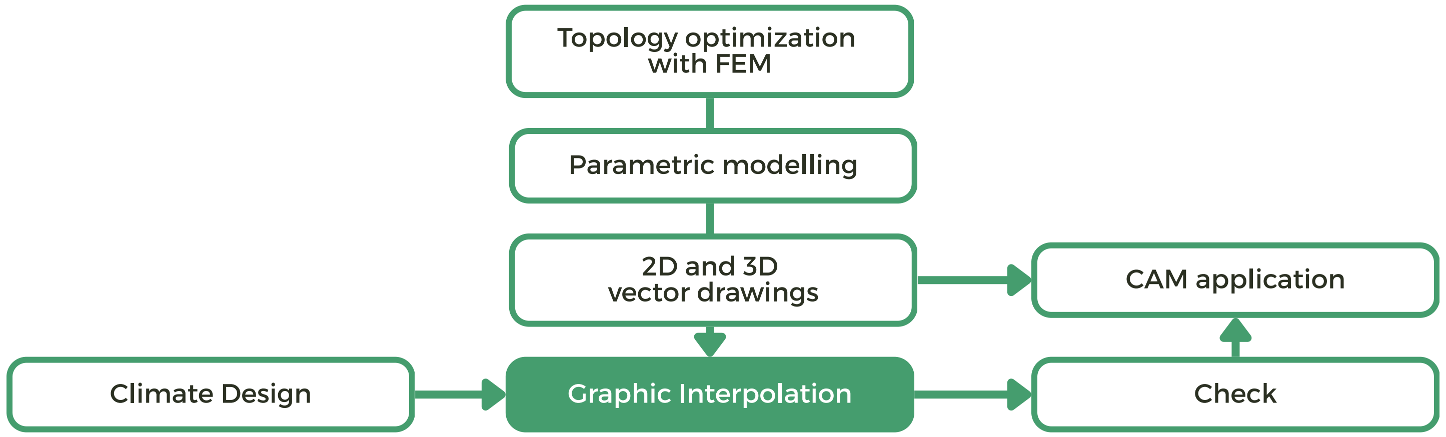
TEAM: Andrea Cutroni

FACULTY: Ilaria Giannetti, Michele Marino

CONTEXT: Research scholarship “Topology Optimization for Architectural Structures: An Ease-to-Use Working Space for Modeling and Design”

The project focuses on the parametric modelling of topology optimized geometries. The Grasshopper code is developed to perform graphic interpolation between polylines, exploring design solutions between two limit cases obtained from finite element analysis. Resulting geometries are verified via FEM analyses and rapid prototyping. A pavilion is designed to explore the possibilities enabled by the graphic interpolation method, combining topology optimization and shading criteria to create a highly efficient structure. 3D-printed and laser-cut prototypes were produced, and the project was presented at Maker Faire 2023 in Rome.

WORKFLOW

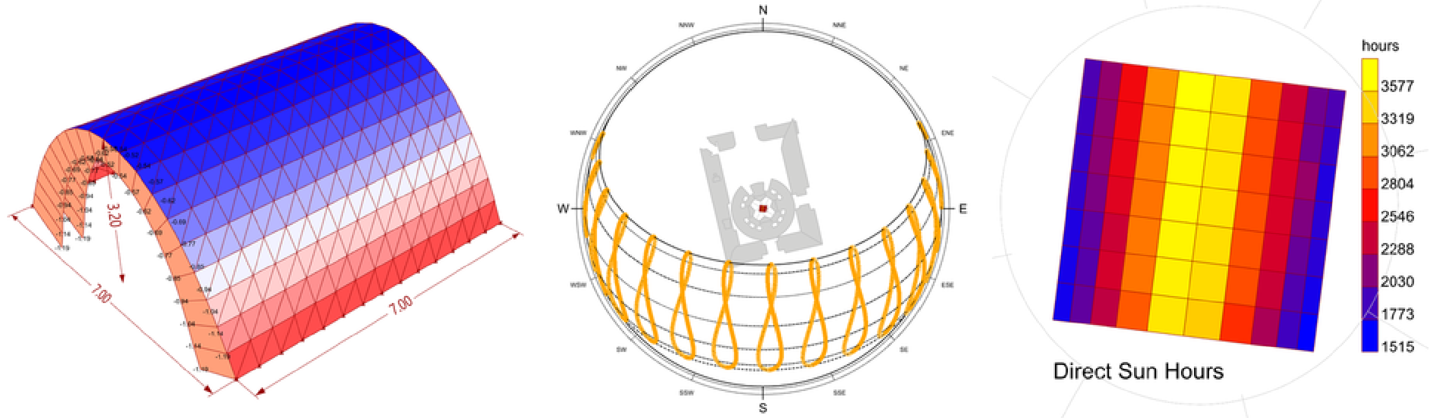


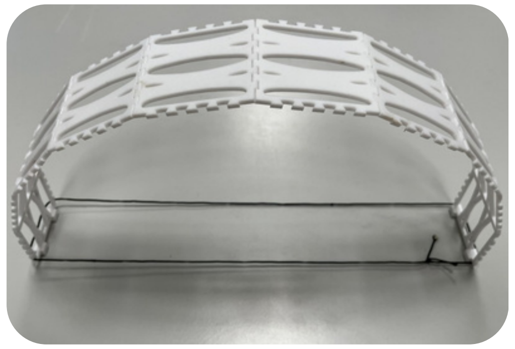
TOPOLOGY OPTIMIZATION & GRAPHIC INTERPOLATION

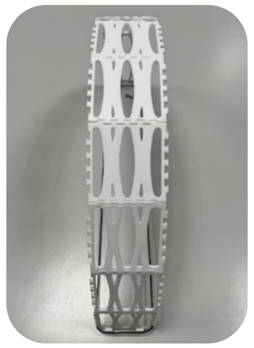
Topology optimization solutions for two limit cases are obtained using the SIMP approach, with the solutions field in between generated through the parametric script.

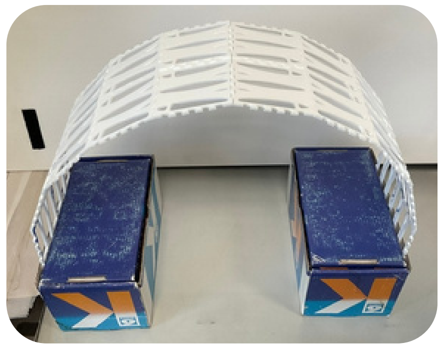
STRUCTURAL & ENVIRONMENTAL ANALYSIS

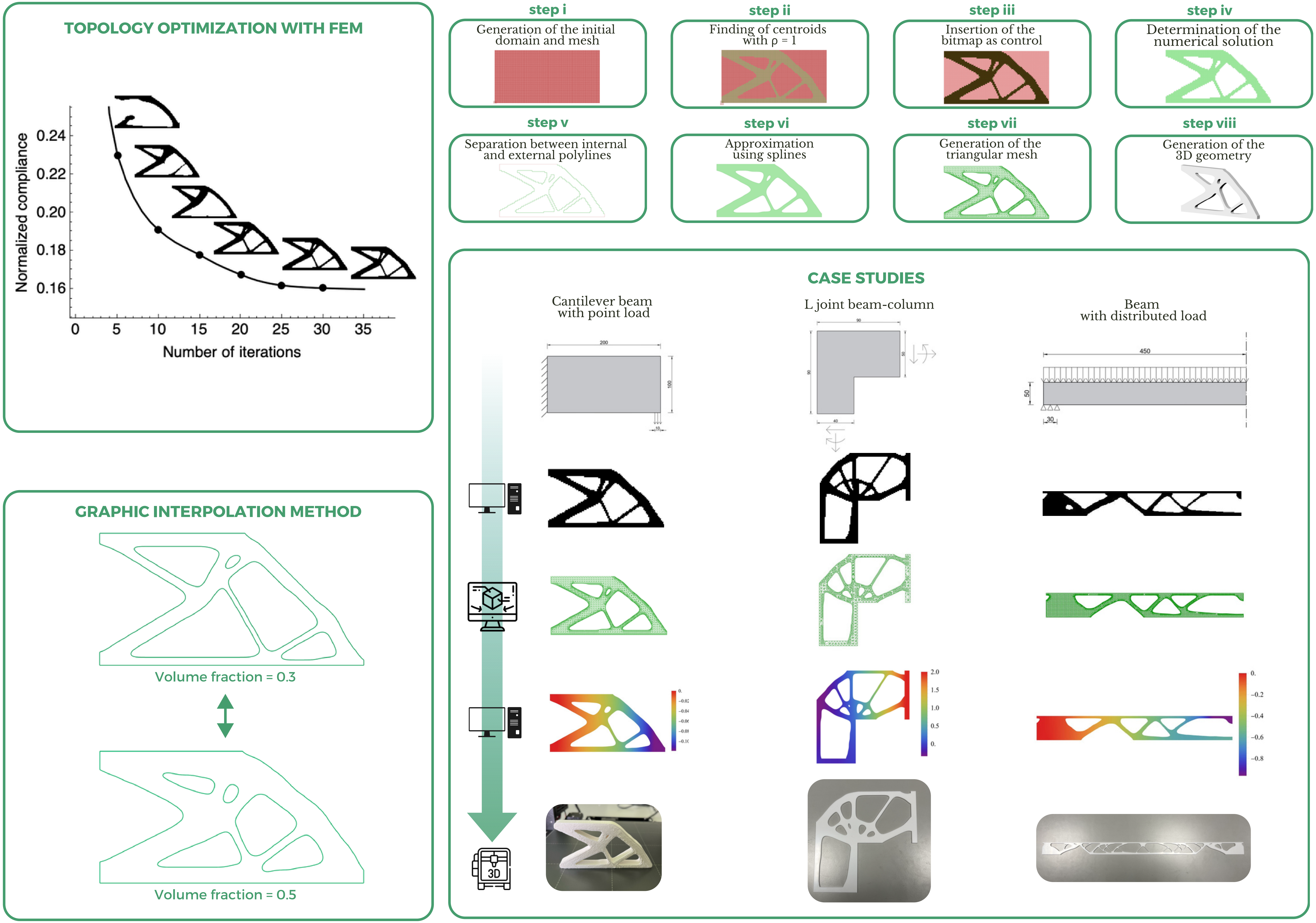
The geometry is based on the catenary shape. Structural analysis is conducted using the Karamba3D plug-in to identify the stress distribution in the pavilion.
The environmental analysis is carried out using the Ladybug plug-in to determine the optimal placement of the panels.











TOPOLOGY OPTIMIZATION WITH FEM
Normalized compliance
0.24
0.22
0.20
0.18
0.16
0
5
10
15
20
25
30
35
Number of iterations
GRAPHIC INTERPOLATION METHOD
Volume fraction = 0.3
Volume fraction = 0.5
step i
Generation of the initial domain and mesh
step ii
Finding of centroids with ρ = 1
step iii
Insertion of the bitmap as control
step iv
Determination of the numerical solution
step v
Separation between internal and external polylines
step vi
Approximation using splines
step vii
Generation of the triangular mesh
step viii
Generation of the 3D geometry
CASE STUDIES
Cantilever beam with point load
L joint beam-column
Beam with distributed load
200
100
90
50
40
450
30
3D
2.0
1.5
1.0
0.5
0.
-0.2
-0.4
-0.6
-0.8